Robust Point Set Registration Using Gaussian Mixture Models

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Introduction

- point set을 **Gaussian Mixture Model**로 표현해 robust하게 registration하는 방법제안
- Registration 문제를 **2개의 Gaussian Mixture Model의 alignment 문제로 재정의**
- **closed-form L2 distance**를 사용해 robust하고 계산 가능한 framwork 제안

Point Set as Gaussian Mixture Model

- Point set을 point list가 아닌 Gaussian Mixture density로 표현
- 일반적인 GMM

$$p(\mathbf{x}) = \sum_{i=1}^{k} w_i \phi(\mathbf{x}|\mu_i, \Sigma_i)$$

$$\phi(\mathbf{x}|\mu_i, \Sigma_i) = \frac{\exp\left[-\frac{1}{2}(\mathbf{x}-\mu_i)^T \Sigma_i^{-1} (\mathbf{x}-\mu_i)\right]}{\sqrt{(2\pi)^d |\det(\Sigma_i)|}}$$

  - k : componet의 수(점 개수)
  - μ_i : component의 mean (점 위치)
  - Σ_i : component의 covariance (점 분포)
  - w_i : component의 weight (점 중요도)

Registration as Density Alignment

- Objective function

$$d_{L_2}(S, \mathcal{M}, \theta) = \int (gmm(S) - gmm(T(\mathcal{M}, \theta)))^2 dx$$

  - S: fixed scene / target point set
  - $\mathcal{M}$: moving model / source point set
  - θ: transformation parameters
  - $T(\mathcal{M}, \theta)$: transformed model set

- Registration을 point residual minimization이 아니라 density discrepancy minimization으로 바꾼다

L2 Distance(Robust + Closed form)

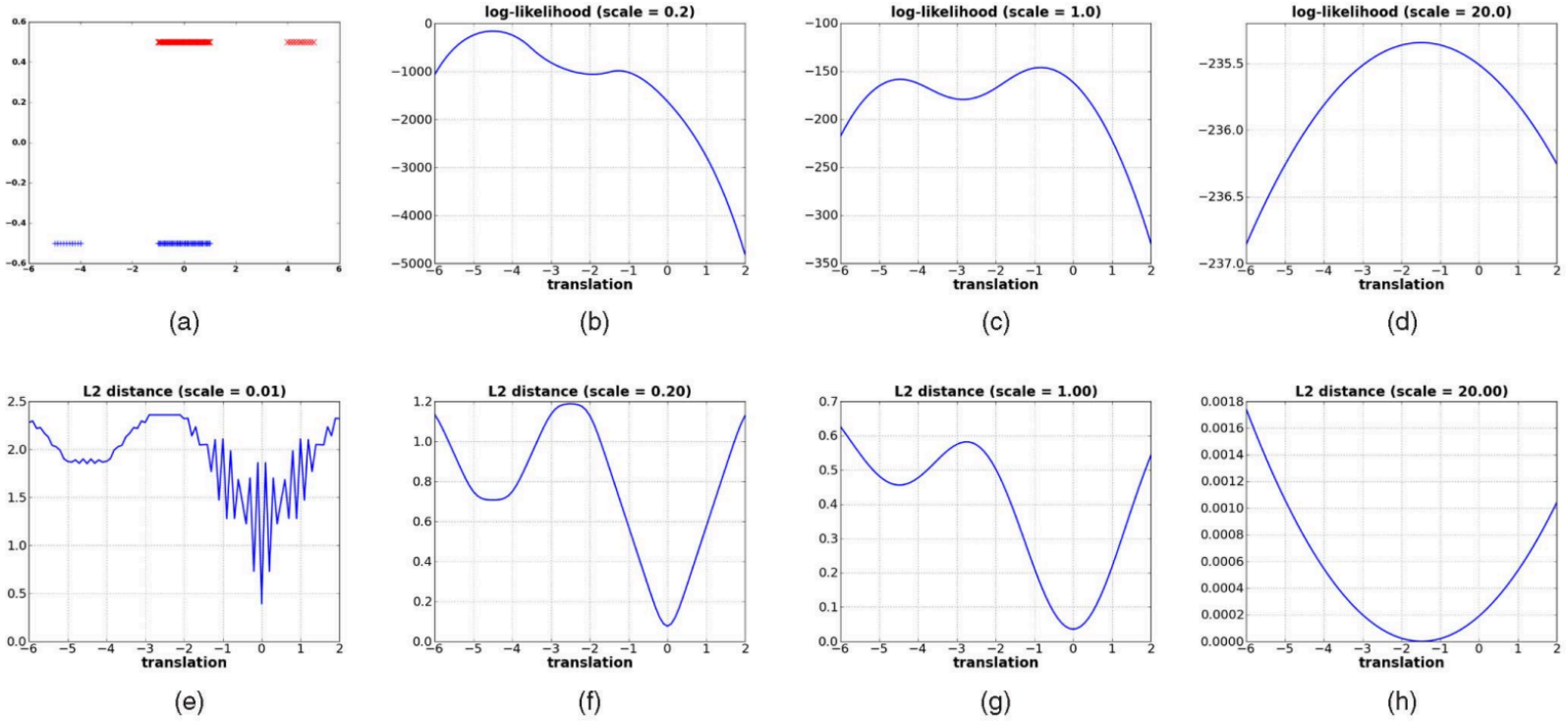

Fig. 2. (a) Two 1D point sets, each of which is clustered into two separate regions of different sampling rates. (b)-(h) The log-likelihood functions and the L2 distances at different scales versus translation of one point set.

- outlier / density 불균형 상황에서 L2와 Log-likelihood에서 더 안정적임

L2 Distance(Robust + Closed form)

- density power divergence

$$d_\alpha(g, f) = \int \left\{ \frac{1}{\alpha} g^{1+\alpha} - \frac{1+\alpha}{\alpha} g f^\alpha + f^{1+\alpha} \right\} dx$$

  - α : 0이면 KL divergence, 1이면 L2 distance

- Gaussian Product Integral

$$\int \phi(\mathbf{x}|\mu_1, \Sigma_1)\phi(\mathbf{x}|\mu_2, \Sigma_2)\, dx = \phi(0|\mu_1 - \mu_2, \Sigma_1 + \Sigma_2)$$

  - Gaussian mixture 사이의 L2 distance가 실제로 계산 가능한 objective function이 됨

Unified Interpretation

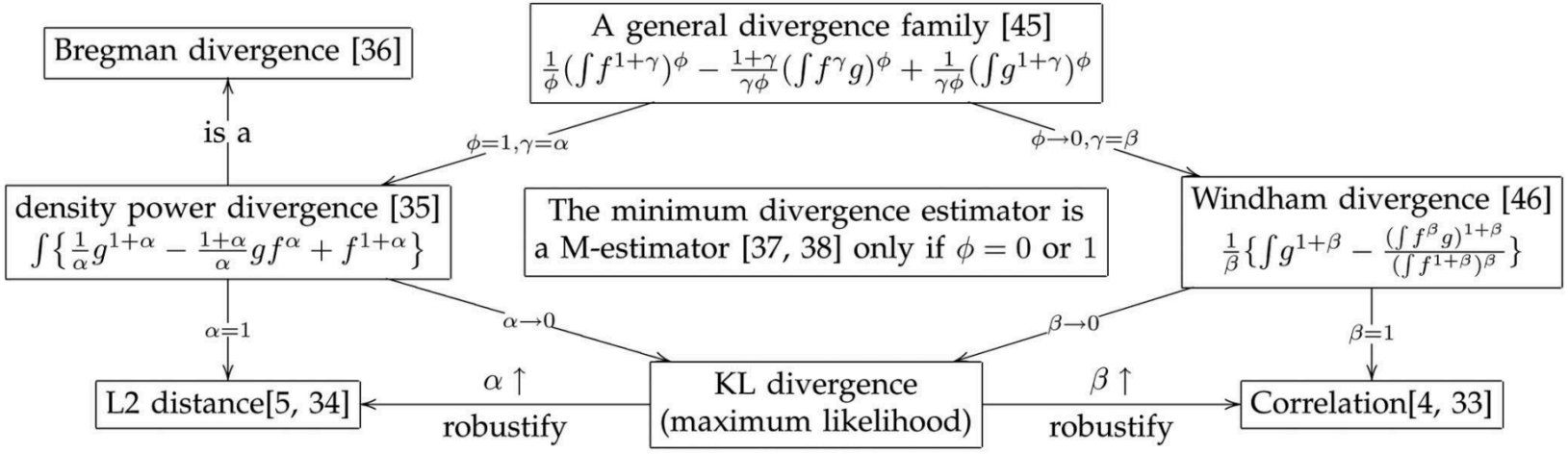

Fig. 3. The relationship between various divergence functions discussed in the text.

- ICP → approximate KL
- EM / MLE → likelihood / KL
- KC → correlation / L2 cross-term
- Proposed → L2 distance between Gaussian mixtures

Practical Algorithms

- Rigid registration
  - translation은 closed-form update
  - rotation은 gradient-based optimization
- Non-rigid registration
  - smooth deformation field 최적화
  - regularization으로 motion coherence 유지

$$\frac{\partial F}{\partial t} = \mathbf{G}^T \mathbf{1}_m$$

$$\frac{\partial F}{\partial r_i} = \mathbf{1}_d^T \left((\mathbf{G}^T \mathbf{M}_0) \otimes \frac{\partial \mathbf{R}}{\partial r_i} \right) \mathbf{1}_d$$

Algorithm 1. A robust algorithm for point set registration using Gaussian mixtures

Input: The *model* set $\mathcal{M}$, the *scene* set $\mathcal{S}$, and a parametrized transformation model T.

Output: The optimal transformation parameter θ of model T that best aligns $\mathcal{M}$ and $\mathcal{S}$.

```
begin
    Estimate an initial scale σ from input point sets;
    Specify an initial parameter θ, e.g., from the identity transform;
    repeat
        Set up the objective function f(θ) as the L2 distance between the Gaussian mixtures constructed from the transformed model T(M, θ) and the scene S with a scale σ. A regularization term can be added depending on the transformation model;
        Optimize the objective function f using a numerical optimization engine (e.g., quasi-Newton algorithm when ∇f is available) with θ as the initial parameter;
        Update the parameter θ ← arg min_T f;
        Decrease the scale σ accordingly as an annealing step;
        untill some stopping criterion is satisfied
end
```

Constraint / Limitation

Constraints

- Density-based registration이므로 sampling density가 비슷한 두 point set에서 더 유리함
- GMM construction은 equal weights, point-centered means, shared spherical covariance를 가정
- 입력은 mesh, label, descriptor 없이 point coordinates만 있는 unstructured point set가 전제

Limitations

- L2 objective가 robust하더라도 local minima는 여전히 남음
- symmetry, repeated patterns, large motion, poor initialization에서는 optimization이 local minimum에 갇힐 수 있음
- 실제 성능은 large-scale start, annealing, multiple starts, coarse-to-fine 같은 heuristic에 크게 의존함
- 모든 registration 문제에서 항상 최선의 방법이라고 보기는 어려움

Conclusion

- Point set registration을 **Gaussian mixture alignment** 문제로 재정의
- **closed-form L2 distance** 기반 objective로 robust하고 계산 가능한 정합 수행
- ICP, EM/MLE, kernel correlation을 **하나의 divergence framework**에서 통합적으로 해석